Kinora Foot

Clinical AI Orientation (Part 11)

Structured reference for RAG / AI-assisted physiotherapy consultation

Bones, Joints, Ligaments, Intrinsics/Extrinsics, Plantar Fascia, Arches, Biomechanics, Neurovascular, Assessment, Pathologies & Rehabilitation

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Disclaimer

Educational Kinora AI resource — not a substitute for licensed clinical care. Synthesis of Gray's/Standring, Moore, Netter, Neumann, Magee, Brukner & Khan, and foot/ankle specialty themes. ***Red flags — suspected Charcot arthropathy or infected diabetic foot, acute Lisfranc injury, open fracture, critical ischemia, or progressive neurologic cavus — require urgent specialist referral.***

Table of Contents

1. Bones

RECORD: First Metatarsal

ID: FOOT-BONE-01

Bone: First Metatarsal

Landmarks: Base, shaft, head; plantar crista for sesamoids

Articulations: Medial cuneiform (1st TMT), proximal phalanx (1st MTP), sesamoids

Muscle Attachments: TA, PL at base; FHB/AH/AdH via sesamoids/plantar plate complex

Ligament Attachments: TMT, MTP collaterals, plantar plate, deep transverse MT ligament

Blood Supply: First dorsal/plantar metatarsal arteries — AVN risk after osteotomy if disrupted

Innervation: Deep peroneal, medial plantar

Ossification: Primary shaft; epiphysis proximal (unique among MTs — base)

Biomechanics: Medial column lever; windlass via sesamoids; high load push-off

Clinical Importance: Hallux valgus/rigidus surgery; bunion; stress rare vs 2-5; sesamoid disorders

Imaging: WB AP/lateral/sesamoid view

References: Standring S. Gray's Anatomy. Moore KL et al. Clinically Oriented Anatomy. Netter FH. Neumann DA. Kinesiology of the Musculoskeletal System. Magee DJ. Orthopedic Physical Assessment. Brukner & Khan Clinical Sports Medicine.

RECORD: Second Metatarsal

ID: FOOT-BONE-02

Bone: Second Metatarsal

Landmarks: Longest MT; recessed base in cuneiform mortise

Articulations: Intermediate cuneiform; medial/lateral cuneiform partial; proximal phalanx

Muscle Attachments: Interossei, adductor hallucis oblique related

Ligament Attachments: Lisfranc ligament (medial cuneiform → 2nd MT base) critical

Blood Supply: Dorsal/plantar MT arteries

Innervation: Deep peroneal, plantar

Ossification: Epiphysis distal

Biomechanics: Keystone midfoot stability; high stress fracture rate

Clinical Importance: Lisfranc; march fracture classic; transfer metatarsalgia

Imaging: WB XR; MRI stress; CT Lisfranc

References: Standring S. Gray's Anatomy. Moore KL et al. Clinically Oriented Anatomy. Netter FH. Neumann DA. Kinesiology of the Musculoskeletal System. Magee DJ. Orthopedic Physical Assessment. Brukner & Khan Clinical Sports Medicine.

RECORD: Third Metatarsal

ID: FOOT-BONE-03

Bone: Third Metatarsal

Landmarks: Base articulates lateral cuneiform

Articulations: Lateral cuneiform, 2nd/4th MT bases, proximal phalanx

Muscle Attachments: Interossei

Ligament Attachments: TMT/intermetatarsal

Blood Supply: MT arteries

Innervation: Deep peroneal/plantar

Ossification: Distal epiphysis

Biomechanics: Central forefoot load share

Clinical Importance: Stress fracture; metatarsalgia

Imaging: WB XR; MRI

References: Standring S. Gray's Anatomy. Moore KL et al. Clinically Oriented Anatomy. Netter FH. Neumann DA. Kinesiology of the Musculoskeletal System. Magee DJ. Orthopedic Physical Assessment. Brukner & Khan Clinical Sports Medicine.

RECORD: Fourth Metatarsal

ID: FOOT-BONE-04

Bone: Fourth Metatarsal

Landmarks: Base with cuboid

Articulations: Cuboid, 3rd/5th MT, proximal phalanx

Muscle Attachments: Interossei

Ligament Attachments: TMT

Blood Supply: MT arteries

Innervation: Superficial/deep peroneal, lateral plantar

Ossification: Distal epiphysis

Biomechanics: Lateral forefoot

Clinical Importance: Stress fracture; Freiberg rare

Imaging: WB XR

References: Standring S. Gray's Anatomy. Moore KL et al. Clinically Oriented Anatomy. Netter FH. Neumann DA. Kinesiology of the Musculoskeletal System. Magee DJ. Orthopedic Physical Assessment. Brukner & Khan Clinical Sports Medicine.

RECORD: Fifth Metatarsal

ID: FOOT-BONE-05

Bone: Fifth Metatarsal

Landmarks: Tuberosity (PB insert); metaphyseal-diaphyseal junction (Jones zone); distal shaft

Articulations: Cuboid, 4th MT, proximal phalanx

Muscle Attachments: PB tuberosity; PL groove nearby; ADM; peroneus tertius dorsal base variable

Ligament Attachments: TMT; plantar/dorsal

Blood Supply: Watershed at Jones zone — healing risk

Innervation: Sural, lateral plantar, superficial peroneal

Ossification: Apophysis lateral tuberosity (Iselin) vs avulsion fracture differential in youth

Biomechanics: Lateral column; PB eversion lever

Clinical Importance: Zone 1 avulsion vs Zone 2 Jones vs Zone 3 stress — treatment differs markedly

Imaging: XR; MRI/CT if occult stress

References: Standring S. Gray's Anatomy. Moore KL et al. Clinically Oriented Anatomy. Netter FH. Neumann DA. Kinesiology of the Musculoskeletal System. Magee DJ. Orthopedic Physical Assessment. Brukner & Khan Clinical Sports Medicine.

RECORD: Hallux Phalanges

ID: FOOT-BONE-06

Bone: Hallux Phalanges

Landmarks: Proximal and distal phalanx (IP joint); no middle phalanx

Articulations: 1st MTP; hallux IP

Muscle Attachments: EHL, FHL, FHB, AH, AdH, EHB

Ligament Attachments: Collaterals, plantar plate, sesamoid apparatus

Blood Supply: Plantar/dorsal digital arteries

Innervation: Medial plantar / deep peroneal digital

Ossification: Epiphyses pattern standard

Biomechanics: Windlass and push-off; needs ~60+ deg MTP DF for normal gait push-off teaching values

Clinical Importance: Hallux valgus/rigidus, turf toe, sesamoiditis

Imaging: WB AP/lateral; sesamoid views

References: Standring S. Gray's Anatomy. Moore KL et al. Clinically Oriented Anatomy. Netter FH. Neumann DA. Kinesiology of the Musculoskeletal System. Magee DJ. Orthopedic Physical Assessment. Brukner & Khan Clinical Sports Medicine.

RECORD: Lesser Toe Phalanges

ID: FOOT-BONE-07

Bone: Lesser Toe Phalanges

Landmarks: Proximal, middle, distal phalanges toes 2-5 (5th middle often fused variants)

Articulations: MTP, PIP, DIP

Muscle Attachments: FDL/FDB, EDL/EDB, interossei, lumbricals

Ligament Attachments: Collaterals, plantar plates

Blood Supply: Digital arteries

Innervation: Plantar/digital nerves

Ossification: Standard phalangeal

Biomechanics: Grip and pressure distribution; deformity alters metatarsalgia

Clinical Importance: Hammer/claw/mallet toe; plantar plate tear (crossover toe)

Imaging: XR WB; MRI plantar plate

References: Standring S. Gray's Anatomy. Moore KL et al. Clinically Oriented Anatomy. Netter FH. Neumann DA. Kinesiology of the Musculoskeletal System. Magee DJ. Orthopedic Physical Assessment. Brukner & Khan Clinical Sports Medicine.

RECORD: Hallucal Sesamoids

ID: FOOT-BONE-08

Bone: Hallucal Sesamoids

Landmarks: Tibial (medial) and fibular (lateral) under 1st MT head

Articulations: First MT head crista grooves

Muscle Attachments: FHB tendons embed; AH/AdH attachments via apparatus

Ligament Attachments: Sesamoid ligaments / plantar plate complex

Blood Supply: Plantar arteries — bipartite vs fracture differential

Innervation: Medial plantar branches

Ossification: Often multipartite (esp. tibial) — normal variant vs acute fracture

Biomechanics: Reduce friction; increase FHB MA; windlass contributors

Clinical Importance: Sesamoiditis, fracture, AVN (Renander), turf toe involvement

Imaging: Sesamoid axial view; MRI

References: Standring S. Gray's Anatomy. Moore KL et al. Clinically Oriented Anatomy. Netter FH. Neumann DA. Kinesiology of the Musculoskeletal System. Magee DJ. Orthopedic Physical Assessment. Brukner & Khan Clinical Sports Medicine.

2. Joints

RECORD: First MTP Joint

ID: FOOT-JNT-01

Joint: First metatarsophalangeal joint

Type: Synovial condyloid

Capsule: Reinforced by collaterals and plantar plate/sesamoid complex

Articular Surfaces: 1st MT head with proximal phalanx base + sesamoids plantar

Ligaments: Medial/lateral collaterals, plantar plate, sesamoid ligaments

Blood Supply: First MT arteries

Innervation: Medial plantar, deep peroneal

Arthrokinematics: Roll/glide with DF/PF; sesamoids track under head

Osteokinematics: DF critical for push-off (~60 deg functional target often cited); PF; ABD/ADD

Degrees of Freedom: 2 (DF/PF + ABD/ADD)

Stability: Plantar plate + sesamoid apparatus + dynamic FHL/FHB

Biomechanics: Windlass activation with hallux DF tightens plantar fascia

Clinical Tests: ROM DF/PF, grind (OA), Lachman of toe (plantar plate)

Pain Referral: 1st MTP medial/plantar

Common Pathologies: HV, hallux rigidus, turf toe, sesamoiditis

Rehabilitation: Protect grade III turf toe; restore DF; toe flexor strength; rocker footwear selected

References: Standring S. Gray's Anatomy. Moore KL et al. Clinically Oriented Anatomy. Netter FH. Neumann DA. Kinesiology of the Musculoskeletal System. Magee DJ. Orthopedic Physical Assessment. Brukner & Khan Clinical Sports Medicine.

RECORD: Lesser MTP Joints

ID: FOOT-JNT-02

Joint: Lesser metatarsophalangeal joints (2-5)

Type: Synovial condyloid

Capsule: Collaterals + plantar plates

Articular Surfaces: MT heads with proximal phalanges

Ligaments: Collaterals, plantar plates, deep transverse MT ligament

Blood Supply: MT arteries

Innervation: Plantar digital nerves

Arthrokinematics: DF/PF glides

Osteokinematics: DF/PF; minor ABD/ADD

Degrees of Freedom: 2

Stability: Plantar plate primary plantar restraint

Biomechanics: Pressure share; failure → crossover toe / metatarsalgia

Clinical Tests: Vertical Lachman / drawer of toe; Mulder for neuroma interspace

Pain Referral: Plantar MT heads

Common Pathologies: Plantar plate tear, Freiberg, metatarsalgia, rheumatoid

Rehabilitation: Toe yoga, intrinsic strength, metatarsal pads, taping crossover toe

References: Standring S. Gray's Anatomy. Moore KL et al. Clinically Oriented Anatomy. Netter FH. Neumann DA. Kinesiology of the Musculoskeletal System. Magee DJ. Orthopedic Physical Assessment. Brukner & Khan Clinical Sports Medicine.

RECORD: IP Joints of Toes

ID: FOOT-JNT-03

Joint: Interphalangeal joints (hallux IP; lesser PIP/DIP)

Type: Synovial hinge

Capsule: Collaterals + plantar plate

Articular Surfaces: Phalangeal heads/bases

Ligaments: Collaterals, plantar plates

Blood Supply: Digital arteries

Innervation: Digital nerves

Arthrokinematics: Hinge roll/glide

Osteokinematics: Flexion/extension

Degrees of Freedom: 1

Stability: Collaterals

Biomechanics: Fine grip; deformity sites (hammer/mallet)

Clinical Tests: ROM; flexibility rigid vs flexible hammer toe

Pain Referral: Toe tips/dorsal PIP

Common Pathologies: Hammer/claw/mallet; arthritis

Rehabilitation: Flexible deformities — stretching/strengthening; rigid may need surgery consult

References: Standring S. Gray's Anatomy. Moore KL et al. Clinically Oriented Anatomy. Netter FH. Neumann DA. Kinesiology of the Musculoskeletal System. Magee DJ. Orthopedic Physical Assessment. Brukner & Khan Clinical Sports Medicine.

RECORD: Lisfranc (TMT) Joint Complex

ID: FOOT-JNT-04

Joint: Tarsometatarsal / Lisfranc complex

Type: Synovial plane joints forming midfoot mortise

Capsule: Dorsal thinner than plantar

Articular Surfaces: Cuneiforms/cuboid with MT bases 1-5

Ligaments: Dorsal, plantar, interosseous; Lisfranc ligament (C1→MT2) critical

Blood Supply: Dorsalis pedis deep plantar branch courses near Lisfranc — injury risk

Innervation: Deep peroneal, plantar

Arthrokinematics: Limited glides — stability > mobility

Osteokinematics: Minimal; midfoot rigidity for push-off

Degrees of Freedom: Limited accessory

Stability: Osseous mortise of 2nd MT + Lisfranc ligament

Biomechanics: Transverse arch keystone; failure collapses midfoot

Clinical Tests: Piano key, midfoot squeeze, stress abduction; high index of suspicion after trauma

Pain Referral: Midfoot dorsal

Common Pathologies: Lisfranc sprain/fracture-dislocation — often missed

Rehabilitation: NWB period common if stable sprain; ORIF/arthrodesis if unstable — surgeon protocols

References: Standring S. Gray's Anatomy. Moore KL et al. Clinically Oriented Anatomy. Netter FH. Neumann DA. Kinesiology of the Musculoskeletal System. Magee DJ. Orthopedic Physical Assessment. Brukner & Khan Clinical Sports Medicine.

RECORD: Intermetatarsal Joints

ID: FOOT-JNT-05

Joint: Intermetatarsal articulations / soft-tissue spaces

Type: Synovial plane (bases) + soft-tissue interspaces

Capsule: Limited base capsules

Articular Surfaces: MT bases

Ligaments: Deep transverse metatarsal ligament; interosseous

Blood Supply: MT network

Innervation: Digital/interdigital nerves (Morton neuroma site)

Arthrokinematics: Minimal

Osteokinematics: Splay under load

Degrees of Freedom: Accessory

Stability: Deep transverse MT ligament

Biomechanics: Forefoot splay; neuroma compression in 3-4 interspace classic

Clinical Tests: Mulder's click

Pain Referral: Web space / toes

Common Pathologies: Morton neuroma; bursitis

Rehabilitation: Wide toe box, met pads, injection selected

References: Standring S. Gray's Anatomy. Moore KL et al. Clinically Oriented Anatomy. Netter FH. Neumann DA. Kinesiology of the Musculoskeletal System. Magee DJ. Orthopedic Physical Assessment. Brukner & Khan Clinical Sports Medicine.

3. Ligaments

RECORD: Lisfranc Ligament Complex

ID: FOOT-LIG-01

Ligament: Lisfranc ligament (medial cuneiform to 2nd MT base) + dorsal/plantar TMT ligaments

Origin: Medial cuneiform (Lisfranc proper)

Insertion: Base 2nd MT medial

Fiber Orientation: Oblique plantar-medial strongest component often plantar/interosseous

Function: Prevents MT2 separation from medial column; midfoot stability keystone

Biomechanics: Failure allows diastasis and midfoot collapse under load

Injury Mechanism: Axial load on PF foot; twist; indirect sports trauma — often subtle

Healing: Unstable injuries need surgical stabilization; missed injury → midfoot OA/collapse

Clinical Tests: Pain midfoot, inability to toe raise, diastasis on WB XR

Imaging: WB bilateral AP — look >2 mm diastasis; CT/MRI

Rehabilitation: Strict per stability/surgery; delayed impact

References: Standring S. Gray's Anatomy. Moore KL et al. Clinically Oriented Anatomy. Netter FH. Neumann DA. Kinesiology of the Musculoskeletal System. Magee DJ. Orthopedic Physical Assessment. Brukner & Khan Clinical Sports Medicine.

RECORD: Plantar Plate (MTP)

ID: FOOT-LIG-02

Ligament: Plantar plate of MTP joints

Origin: Metatarsal neck periosteum / plantar capsule

Insertion: Proximal phalanx plantar base

Fiber Orientation: Plantar fibrocartilage plate

Function: Resists MTP hyperextension / dorsal subluxation

Biomechanics: Fails in crossover 2nd toe deformity continuum

Injury Mechanism: Repetitive overload, acute turf-toe like lesser toes

Healing: Partial may stabilize with taping/offload; complete may need repair

Clinical Tests: Vertical drawer, crossover deformity

Imaging: MRI

Rehabilitation: Taping, rocker shoe, intrinsic strength

References: Standring S. Gray's Anatomy. Moore KL et al. Clinically Oriented Anatomy. Netter FH. Neumann DA. Kinesiology of the Musculoskeletal System. Magee DJ. Orthopedic Physical Assessment. Brukner & Khan Clinical Sports Medicine.

RECORD: Deep Transverse Metatarsal Ligament

ID: FOOT-LIG-03

Ligament: Deep transverse metatarsal ligament

Origin: Plantar plate / MT head region spanning

Insertion: Adjacent MT heads/plantar plates

Fiber Orientation: Transverse forefoot

Function: Prevents excessive MT splay; neuroma lies dorsal to it

Biomechanics: Compression corridor for interdigital nerve

Injury Mechanism: Not typically isolated tear focus — neuroma context

Healing: N/A primary

Clinical Tests: Mulder

Imaging: US/MRI neuroma

Rehabilitation: Forefoot offload strategies

References: Standring S. Gray's Anatomy. Moore KL et al. Clinically Oriented Anatomy. Netter FH. Neumann DA. Kinesiology of the Musculoskeletal System. Magee DJ. Orthopedic Physical Assessment. Brukner & Khan Clinical Sports Medicine.

RECORD: Long & Short Plantar Ligaments

ID: FOOT-LIG-04

Ligament: Long plantar and short plantar (plantar calcaneocuboid) ligaments

Origin: Calcaneus plantar

Insertion: Cuboid ± MT bases (long plantar)

Fiber Orientation: Longitudinal lateral column

Function: Static lateral longitudinal arch support

Biomechanics: PL tendon runs in long plantar tunnel/groove region

Injury Mechanism: Midfoot sprain continuum

Healing: With lateral column sprain care

Clinical Tests: Lateral midfoot pain map

Imaging: MRI

Rehabilitation: Load progressive; PL capacity

References: Standring S. Gray's Anatomy. Moore KL et al. Clinically Oriented Anatomy. Netter FH. Neumann DA. Kinesiology of the Musculoskeletal System. Magee DJ. Orthopedic Physical Assessment. Brukner & Khan Clinical Sports Medicine.

RECORD: Toe Collateral Ligaments

ID: FOOT-LIG-05

Ligament: MTP/IP collateral ligaments

Origin: MT/phalanx heads

Insertion: Adjacent phalanx bases

Fiber Orientation: Medial/lateral

Function: Varus/valgus stability of toes

Biomechanics: HV attenuates medial 1st MTP restraints

Injury Mechanism: Trauma; chronic HV

Healing: Taping/stabilize; surgical in HV correction

Clinical Tests: Varus/valgus stress toes

Imaging: XR alignment; MRI selected

Rehabilitation: Protection; alignment footwear

References: Standring S. Gray's Anatomy. Moore KL et al. Clinically Oriented Anatomy. Netter FH. Neumann DA. Kinesiology of the Musculoskeletal System. Magee DJ. Orthopedic Physical Assessment. Brukner & Khan Clinical Sports Medicine.

4. Intrinsic Muscles

RECORD: Abductor Hallucis

ID: FOOT-INT-01

Muscle: Abductor Hallucis

Origin: Calcaneal tuberosity / flexor retinaculum

Insertion: Medial proximal phalanx hallux / medial sesamoid

Innervation: Medial plantar (S2-S3)

Blood Supply: Medial plantar artery

Function: Abducts/flexes hallux; medial arch dynamic

EMG: High in short-foot tasks

Clinical Importance: HV medial support; Baxter nerve nearby

Exercises: Short foot, toe spread, doming

Progressions: Isometrics → walking doming → run

References: Standring S. Gray's Anatomy. Moore KL et al. Clinically Oriented Anatomy. Netter FH. Neumann DA. Kinesiology of the Musculoskeletal System. Magee DJ. Orthopedic Physical Assessment. Brukner & Khan Clinical Sports Medicine.

RECORD: Flexor Digitorum Brevis

ID: FOOT-INT-02

Muscle: Flexor Digitorum Brevis

Origin: Calcaneal tuberosity

Insertion: Middle phalanges 2-5

Innervation: Medial plantar

Blood Supply: Medial plantar artery

Function: Flexes PIP toes 2-5

EMG: Stance grip

Clinical Importance: Hammer toe balance with intrinsics

Exercises: Towel curls, piano toes

Progressions: Add load / single-leg

References: Standring S. Gray's Anatomy. Moore KL et al. Clinically Oriented Anatomy. Netter FH. Neumann DA. Kinesiology of the Musculoskeletal System. Magee DJ. Orthopedic Physical Assessment. Brukner & Khan Clinical Sports Medicine.

RECORD: Abductor Digiti Minimi

ID: FOOT-INT-03

Muscle: Abductor Digiti Minimi

Origin: Calcaneal tuberosity

Insertion: Lateral 5th proximal phalanx

Innervation: Lateral plantar

Blood Supply: Lateral plantar artery

Function: Abducts 5th toe; lateral arch assist

EMG: Stance

Clinical Importance: 5th ray stability

Exercises: 5th toe abduction drills

Progressions: Barefoot control progressions

References: Standring S. Gray's Anatomy. Moore KL et al. Clinically Oriented Anatomy. Netter FH. Neumann DA. Kinesiology of the Musculoskeletal System. Magee DJ. Orthopedic Physical Assessment. Brukner & Khan Clinical Sports Medicine.

RECORD: Quadratus Plantae

ID: FOOT-INT-04

Muscle: Quadratus Plantae

Origin: Calcaneus plantar

Insertion: FDL tendon margin

Innervation: Lateral plantar

Blood Supply: Lateral plantar artery

Function: Straightens FDL pull line

EMG: With FDL

Clinical Importance: Toe flexion efficiency

Exercises: Toe flexion with alignment

Progressions: Combined FDL loading

References: Standring S. Gray's Anatomy. Moore KL et al. Clinically Oriented Anatomy. Netter FH. Neumann DA. Kinesiology of the Musculoskeletal System. Magee DJ. Orthopedic Physical Assessment. Brukner & Khan Clinical Sports Medicine.

RECORD: Lumbricals (Foot)

ID: FOOT-INT-05

Muscle: Lumbricals (Foot)

Origin: FDL tendons

Insertion: Extensor expansions / proximal phalanges

Innervation: 1st medial plantar; 2-4 lateral plantar

Blood Supply: Plantar arteries

Function: Flex MTP / extend IP (balance)

EMG: Fine control

Clinical Importance: Claw toe when intrinsic weak / extrinsic dominant

Exercises: Intrinsic toe yoga

Progressions: Progress to balance tasks

References: Standring S. Gray's Anatomy. Moore KL et al. Clinically Oriented Anatomy. Netter FH. Neumann DA. Kinesiology of the Musculoskeletal System. Magee DJ. Orthopedic Physical Assessment. Brukner & Khan Clinical Sports Medicine.

RECORD: Flexor Hallucis Brevis

ID: FOOT-INT-06

Muscle: Flexor Hallucis Brevis

Origin: Cuboid/lateral cuneiform

Insertion: Both sides proximal phalanx via sesamoids

Innervation: Medial plantar

Blood Supply: Medial plantar artery

Function: Flexes hallux MTP; sesamoid control

EMG: Push-off

Clinical Importance: Sesamoiditis; HV surgery

Exercises: Hallux MTP flexion isometrics

Progressions: Push-off drills

References: Standring S. Gray's Anatomy. Moore KL et al. Clinically Oriented Anatomy. Netter FH. Neumann DA. Kinesiology of the Musculoskeletal System. Magee DJ. Orthopedic Physical Assessment. Brukner & Khan Clinical Sports Medicine.

RECORD: Adductor Hallucis

ID: FOOT-INT-07

Muscle: Adductor Hallucis

Origin: Oblique: MT2-4 bases; Transverse: plantar plates 3-5

Insertion: Lateral hallux proximal phalanx / lateral sesamoid

Innervation: Lateral plantar deep branch

Blood Supply: Plantar arch arteries

Function: Adducts hallux; transverse arch assist

EMG: Forefoot

Clinical Importance: HV (adductor pull lateralizes); transverse arch

Exercises: Toe adduction control, metatarsal dome

Progressions: Footwear + strength

References: Standring S. Gray's Anatomy. Moore KL et al. Clinically Oriented Anatomy. Netter FH. Neumann DA. Kinesiology of the Musculoskeletal System. Magee DJ. Orthopedic Physical Assessment. Brukner & Khan Clinical Sports Medicine.

RECORD: Flexor Digiti Minimi Brevis

ID: FOOT-INT-08

Muscle: Flexor Digiti Minimi Brevis

Origin: Base 5th MT

Insertion: Proximal phalanx 5th

Innervation: Lateral plantar

Blood Supply: Lateral plantar artery

Function: Flexes 5th MTP

EMG: Stance

Clinical Importance: 5th ray

Exercises: 5th toe flexion

Progressions: Barefoot gait

References: Standring S. Gray's Anatomy. Moore KL et al. Clinically Oriented Anatomy. Netter FH. Neumann DA. Kinesiology of the Musculoskeletal System. Magee DJ. Orthopedic Physical Assessment. Brukner & Khan Clinical Sports Medicine.

RECORD: Plantar Interossei

ID: FOOT-INT-09

Muscle: Plantar Interossei

Origin: Medial sides MT3-5

Insertion: Proximal phalanges / extensor expansions

Innervation: Lateral plantar

Blood Supply: Plantar arteries

Function: Adduct toes toward 2nd; flex MTP

EMG: Fine

Clinical Importance: Toe alignment

Exercises: Toe adduction

Progressions: Progress balance

References: Standring S. Gray's Anatomy. Moore KL et al. Clinically Oriented Anatomy. Netter FH. Neumann DA. Kinesiology of the Musculoskeletal System. Magee DJ. Orthopedic Physical Assessment. Brukner & Khan Clinical Sports Medicine.

RECORD: Dorsal Interossei

ID: FOOT-INT-10

Muscle: Dorsal Interossei

Origin: Adjacent MT shafts

Insertion: Proximal phalanges (abduct from 2nd ray axis)

Innervation: Lateral plantar

Blood Supply: Dorsal MT arteries

Function: Abduct toes; flex MTP

EMG: Fine

Clinical Importance: Metatarsalgia space

Exercises: Toe abduction spreads

Progressions: Wide toe box functional

References: Standring S. Gray's Anatomy. Moore KL et al. Clinically Oriented Anatomy. Netter FH. Neumann DA. Kinesiology of the Musculoskeletal System. Magee DJ. Orthopedic Physical Assessment. Brukner & Khan Clinical Sports Medicine.

5. Extrinsic Muscles

RECORD: Tibialis Anterior (foot insertion)

ID: FOOT-EXT-01

Muscle: Tibialis Anterior (foot insertion)

Origin: Lateral tibia

Insertion: Medial cuneiform/1st MT

Innervation: Deep peroneal

Blood Supply: Anterior tibial

Function: DF/inversion; decelerates PF at IC

EMG: IC eccentric

Clinical Importance: Foot drop; first-ray elevation couple with PL

Exercises: Heel walks

Progressions: Gait drills

References: Standring S. Gray's Anatomy. Moore KL et al. Clinically Oriented Anatomy. Netter FH. Neumann DA. Kinesiology of the Musculoskeletal System. Magee DJ. Orthopedic Physical Assessment. Brukner & Khan Clinical Sports Medicine.

RECORD: Tibialis Posterior (foot insertion)

ID: FOOT-EXT-02

Muscle: Tibialis Posterior (foot insertion)

Origin: IO membrane

Insertion: Navicular + plantar inserts

Innervation: Tibial

Blood Supply: PTA

Function: Dynamic arch; inversion

EMG: Midstance

Clinical Importance: PTTD / AAFD

Exercises: Arch control heel rises

Progressions: Stage-based loading

References: Standring S. Gray's Anatomy. Moore KL et al. Clinically Oriented Anatomy. Netter FH. Neumann DA. Kinesiology of the Musculoskeletal System. Magee DJ. Orthopedic Physical Assessment. Brukner & Khan Clinical Sports Medicine.

RECORD: Peroneus Longus (foot)

ID: FOOT-EXT-03

Muscle: Peroneus Longus (foot)

Origin: Fibula

Insertion: Medial cuneiform/1st MT plantar

Innervation: Superficial peroneal

Blood Supply: Fibular

Function: PF 1st ray; eversion

EMG: Push-off/lateral

Clinical Importance: 1st ray stability vs TA

Exercises: PL heel raises

Progressions: Uneven ground

References: Standring S. Gray's Anatomy. Moore KL et al. Clinically Oriented Anatomy. Netter FH. Neumann DA. Kinesiology of the Musculoskeletal System. Magee DJ. Orthopedic Physical Assessment. Brukner & Khan Clinical Sports Medicine.

RECORD: Peroneus Brevis (foot)

ID: FOOT-EXT-04

Muscle: Peroneus Brevis (foot)

Origin: Fibula

Insertion: 5th MT tuberosity

Innervation: Superficial peroneal

Blood Supply: Fibular

Function: Eversion

EMG: Lateral ankle

Clinical Importance: 5th MT avulsion / tears

Exercises: Eversion band

Progressions: CAI program

References: Standring S. Gray's Anatomy. Moore KL et al. Clinically Oriented Anatomy. Netter FH. Neumann DA. Kinesiology of the Musculoskeletal System. Magee DJ. Orthopedic Physical Assessment. Brukner & Khan Clinical Sports Medicine.

RECORD: FHL (foot)

ID: FOOT-EXT-05

Muscle: FHL (foot)

Origin: Fibula

Insertion: Hallux distal phalanx

Innervation: Tibial

Blood Supply: Fibular/PTA

Function: Hallux PF push-off

EMG: Toe-off

Clinical Importance: Turf toe continuum / sesamoid load

Exercises: Toe flexion calf raises

Progressions: Push-off plyos

References: Standring S. Gray's Anatomy. Moore KL et al. Clinically Oriented Anatomy. Netter FH. Neumann DA. Kinesiology of the Musculoskeletal System. Magee DJ. Orthopedic Physical Assessment. Brukner & Khan Clinical Sports Medicine.

RECORD: FDL (foot)

ID: FOOT-EXT-06

Muscle: FDL (foot)

Origin: Tibia

Insertion: Distal phalanges 2-5

Innervation: Tibial

Blood Supply: PTA

Function: Toe grip

EMG: Stance

Clinical Importance: Claw toe if overactive vs intrinsics

Exercises: Towel curls with MTP control

Progressions: Intrinsic balance

References: Standring S. Gray's Anatomy. Moore KL et al. Clinically Oriented Anatomy. Netter FH. Neumann DA. Kinesiology of the Musculoskeletal System. Magee DJ. Orthopedic Physical Assessment. Brukner & Khan Clinical Sports Medicine.

RECORD: EHL/EDL (foot)

ID: FOOT-EXT-07

Muscle: EHL/EDL (foot)

Origin: Leg

Insertion: Distal phalanx hallux / toes 2-5

Innervation: Deep peroneal

Blood Supply: Anterior tibial

Function: Toe clearance

EMG: Swing

Clinical Importance: Laceration; L5 testing

Exercises: Toe extension

Progressions: Clearance drills

References: Standring S. Gray's Anatomy. Moore KL et al. Clinically Oriented Anatomy. Netter FH. Neumann DA. Kinesiology of the Musculoskeletal System. Magee DJ. Orthopedic Physical Assessment. Brukner & Khan Clinical Sports Medicine.

RECORD: Achilles (foot insertion)

ID: FOOT-EXT-08

Muscle: Achilles (foot insertion)

Origin: Triceps surae

Insertion: Calcaneal tuberosity

Innervation: Tibial

Blood Supply: Watershed mid-tendon

Function: PF power into foot lever

EMG: Propulsion

Clinical Importance: Insertional vs midportion pathology; Haglund

Exercises: Heel raises per tendinopathy rules

Progressions: Energy storage late

References: Standring S. Gray's Anatomy. Moore KL et al. Clinically Oriented Anatomy. Netter FH. Neumann DA. Kinesiology of the Musculoskeletal System. Magee DJ. Orthopedic Physical Assessment. Brukner & Khan Clinical Sports Medicine.

6. Plantar Fascia

RECORD: Plantar Fascia (Aponeurosis)

ID: FOOT-PF-01

Structure: Dense plantar aponeurosis — central, medial, lateral bands

Attachments: Medial calcaneal tuberosity → digitate slips blending with plantar plates / proximal phalanges via windlass

Continuity: Continuous with Achilles/calcaneal periosteum continuum conceptually (superficial posterior chain)

Windlass Mechanism: Hallux DF winds fascia around MT heads → raises MLA and stiffens foot for push-off (Hicks)

Biomechanics: Static arch support + dynamic windlass; stores/releases energy in running

Clinical Importance: Plantar fasciopathy most common cause of inferior heel pain in runners/adults

Assessment: First-step pain, medial calcaneal tubercle tenderness, windlass test, exclude fat pad/nerve/stress Fx

Treatment: Load management, plantar fascia–specific stretching, calf loading, orthoses selected, CSI limited/ judicious, shockwave selected evidence

Rehabilitation: Pain-guided loading; avoid sudden hill/sprint spikes; footwear review; night splints optional adjunct

References: Standring S. Gray's Anatomy. Moore KL et al. Clinically Oriented Anatomy. Netter FH. Neumann DA. Kinesiology of the Musculoskeletal System. Magee DJ. Orthopedic Physical Assessment. Brukner & Khan Clinical Sports Medicine. Hicks windlass; plantar fasciopathy clinical practice guidelines themes.

7. Arches

RECORD: Medial Longitudinal Arch

ID: FOOT-ARCH-01

Arch: Medial longitudinal arch (MLA)

Bones: Calcaneus, talus, navicular, cuneiforms, MT1-3

Ligament Support: Spring ligament, plantar fascia, deltoid/TN complex

Muscular Support: TP primary dynamic; intrinsics; FHL/FDL; TA

Biomechanics: Shock absorption early stance; windlass stiffens late stance

Clinical Importance: Pes planus / PTTD; pes cavus high arch overload lateral/forefoot

Pathologies: AAFD, plantar fasciopathy tension, navicular stress

Assessment: Navicular drop/drift, Feiss line, too-many-toes, WB XR Meary angle

Rehabilitation: TP/intrinsics, orthoses stage-based, footwear

References: Standring S. Gray's Anatomy. Moore KL et al. Clinically Oriented Anatomy. Netter FH. Neumann DA. Kinesiology of the Musculoskeletal System. Magee DJ. Orthopedic Physical Assessment. Brukner & Khan Clinical Sports Medicine.

RECORD: Lateral Longitudinal Arch

ID: FOOT-ARCH-02

Arch: Lateral longitudinal arch

Bones: Calcaneus, cuboid, MT4-5

Ligament Support: Long/short plantar ligaments, lateral plantar fascia band

Muscular Support: PL, PB, ADM

Biomechanics: Lower/stabler than MLA; contact in stance

Clinical Importance: Cuboid syndrome; 5th MT overload in cavovarus

Pathologies: Jones fracture risk mechanics

Assessment: Lateral column pain map

Rehabilitation: PL capacity; lateral column load management

References: Standring S. Gray's Anatomy. Moore KL et al. Clinically Oriented Anatomy. Netter FH. Neumann DA. Kinesiology of the Musculoskeletal System. Magee DJ. Orthopedic Physical Assessment. Brukner & Khan Clinical Sports Medicine.

RECORD: Transverse Arch

ID: FOOT-ARCH-03

Arch: Transverse metatarsal arch

Bones: Cuneiforms/cuboid midfoot + MT heads forefoot

Ligament Support: Deep transverse MT ligament; interossei

Muscular Support: Adductor hallucis transverse; PL

Biomechanics: Forefoot splay control; pressure distribution

Clinical Importance: Metatarsalgia; Morton neuroma; collapsed forefoot splay

Pathologies: Transfer lesions after 1st ray insufficiency

Assessment: Callus pattern; Mulder; metatarsal pad trial

Rehabilitation: Intrinsic dome, met pads, shoe width

References: Standring S. Gray's Anatomy. Moore KL et al. Clinically Oriented Anatomy. Netter FH. Neumann DA. Kinesiology of the Musculoskeletal System. Magee DJ. Orthopedic Physical Assessment. Brukner & Khan Clinical Sports Medicine.

8. Biomechanics

RECORD: Windlass Mechanism

ID: FOOT-BIO-01

Topic: Windlass

Description: Hallux DF tightens plantar fascia around MT heads elevating MLA (Hicks)

Clinical Relevance: Limited hallux DF impairs push-off stiffness; fasciopathy pain with windlass provocation

Exercises: Great toe DF mobility if limited; controlled heel rises

References: Standring S. Gray's Anatomy. Moore KL et al. Clinically Oriented Anatomy. Netter FH. Neumann DA. Kinesiology of the Musculoskeletal System. Magee DJ. Orthopedic Physical Assessment. Brukner & Khan Clinical Sports Medicine.

RECORD: Push-off

ID: FOOT-BIO-02

Topic: Push-off / propulsion

Description: PF power via Achilles + windlass + hallux DF + FHL

Clinical Relevance: Hallux rigidus/turf toe reduce efficient push-off

Exercises: Heel rises, hallux DF strength, plyometrics graded

References: Standring S. Gray's Anatomy. Moore KL et al. Clinically Oriented Anatomy. Netter FH. Neumann DA. Kinesiology of the Musculoskeletal System. Magee DJ. Orthopedic Physical Assessment. Brukner & Khan Clinical Sports Medicine.

RECORD: Foot Pronation/Supination

ID: FOOT-BIO-03

Topic: Pronation and supination of the foot

Description: Triplanar STJ+midtarsal motion; closed-chain tibial coupling

Clinical Relevance: Neither maximally pronated nor rigid is inherently pathologic — capacity and symptoms matter

Exercises: TP control, cadence, footwear matching load

References: Standring S. Gray's Anatomy. Moore KL et al. Clinically Oriented Anatomy. Netter FH. Neumann DA. Kinesiology of the Musculoskeletal System. Magee DJ. Orthopedic Physical Assessment. Brukner & Khan Clinical Sports Medicine.

RECORD: Load Distribution

ID: FOOT-BIO-04

Topic: Plantar load distribution

Description: Heel → lateral border → metatarsal heads → hallux in typical walking pressure progression

Clinical Relevance: Callus maps overload; diabetic offloading critical

Exercises: Gait training, orthoses, metatarsal doming

References: Standring S. Gray's Anatomy. Moore KL et al. Clinically Oriented Anatomy. Netter FH. Neumann DA. Kinesiology of the Musculoskeletal System. Magee DJ. Orthopedic Physical Assessment. Brukner & Khan Clinical Sports Medicine.

RECORD: Walking/Running/Cutting Foot Demands

ID: FOOT-BIO-05

Topic: Locomotor foot demands

Description: Walking moderate loads; running multiplies pressure/rates; cutting adds shear INV/EV

Clinical Relevance: BSI, neuroma, fasciopathy flare with spikes

Exercises: Graded exposure, surface progression

References: Standring S. Gray's Anatomy. Moore KL et al. Clinically Oriented Anatomy. Netter FH. Neumann DA. Kinesiology of the Musculoskeletal System. Magee DJ. Orthopedic Physical Assessment. Brukner & Khan Clinical Sports Medicine.

9. Neurovascular

RECORD: Medial Plantar Nerve

ID: FOOT-NV-01

Nerve: Medial plantar nerve

Sensory Distribution: Medial sole / medial 3.5 toes plantar

Motor Supply: AH, FDB, FHB, 1st lumbrical

Entrapment Sites: Jogger's foot — entrapment in tunnel near navicular

Clinical Tests: Tinel medial arch; sensory map

Vascular Supply: Travels with medial plantar artery

Pulses: Document PT/DP

References: Standring S. Gray's Anatomy. Moore KL et al. Clinically Oriented Anatomy. Netter FH. Neumann DA. Kinesiology of the Musculoskeletal System. Magee DJ. Orthopedic Physical Assessment. Brukner & Khan Clinical Sports Medicine.

RECORD: Lateral Plantar Nerve / Baxter

ID: FOOT-NV-02

Nerve: Lateral plantar nerve incl. Baxter's nerve (1st branch)

Sensory Distribution: Lateral sole / lateral 1.5 toes; Baxter mainly motor to ADM ± sensory calcaneal contribution variable

Motor Supply: ADM, QP, AddH, FDMB, interossei, lumbricals 2-4

Entrapment Sites: Baxter nerve under AH — chronic medial heel pain differential vs fasciopathy

Clinical Tests: ADM weakness rare clinically; pain map; diagnostic US/MRI selected

Vascular Supply: Lateral plantar artery

Pulses: PT pulse proximal

References: Standring S. Gray's Anatomy. Moore KL et al. Clinically Oriented Anatomy. Netter FH. Neumann DA. Kinesiology of the Musculoskeletal System. Magee DJ. Orthopedic Physical Assessment. Brukner & Khan Clinical Sports Medicine.

RECORD: Interdigital Nerves (Morton)

ID: FOOT-NV-03

Nerve: Interdigital plantar nerves (esp. 3-4 web)

Sensory Distribution: Adjacent toe sides

Motor Supply: None significant clinically

Entrapment Sites: Under deep transverse MT ligament — Morton neuroma

Clinical Tests: Mulder's click; web space squeeze pain

Vascular Supply: Digital arteries

Pulses: DP/PT limb context

References: Standring S. Gray's Anatomy. Moore KL et al. Clinically Oriented Anatomy. Netter FH. Neumann DA. Kinesiology of the Musculoskeletal System. Magee DJ. Orthopedic Physical Assessment. Brukner & Khan Clinical Sports Medicine.

RECORD: Dorsalis Pedis / PT Pulses

ID: FOOT-NV-04

Nerve: Vascular assessment landmark record

Sensory Distribution: N/A

Motor Supply: N/A

Entrapment Sites: PAD / trauma / compartment — ischemia emergency

Clinical Tests: Palpate DP (lat to EHL) and PT (posteroinferior medial malleolus); capillary refill; ABI if indicated

Vascular Supply: DP continuation of anterior tibial; PT → medial/lateral plantar arch

Pulses: DP + PT mandatory after trauma and in diabetics

References: Standring S. Gray's Anatomy. Moore KL et al. Clinically Oriented Anatomy. Netter FH. Neumann DA. Kinesiology of the Musculoskeletal System. Magee DJ. Orthopedic Physical Assessment. Brukner & Khan Clinical Sports Medicine.

10. Functional Assessment

RECORD: Foot Functional Assessment

ID: FOOT-FUNC-01

ROM: 1st MTP DF/PF, ankle DF, hallux IP; STJ INV/EV

Strength: Intrinsics (doming), TP heel rise, toe flexion, FHL

Gait: Heel-to-toe, early heel-off, propulsive hallux, callus pattern

Balance: SLS, foam, YBT

Functional Tests: Heel rises, single-leg hops, step-downs

Hop Tests: As ankle battery when RTS foot injuries

Heel Raise: Inversion for TP; height symmetry

Landing: Forefoot control without excessive INV

Return-to-Sport Criteria: Pain-free push-off, hop LSI, sport volume tolerance, footwear plan

Outcome Measures: FAAM, FFI, FAOS, VISA-A if Achilles related

References: Standring S. Gray's Anatomy. Moore KL et al. Clinically Oriented Anatomy. Netter FH. Neumann DA. Kinesiology of the Musculoskeletal System. Magee DJ. Orthopedic Physical Assessment. Brukner & Khan Clinical Sports Medicine.

11. Special Tests

RECORD: Windlass Test

ID: FOOT-TEST-01

Test: Windlass test

Purpose: Provocation for plantar fasciopathy

Positive Finding: Heel pain reproduced with passive hallux DF WB/NWB variants

Sensitivity: Moderate/variable

Specificity: Higher specificity reported in some studies vs Sn

Evidence: Useful cluster with history of first-step pain — not standalone

References: Standring S. Gray's Anatomy. Moore KL et al. Clinically Oriented Anatomy. Netter FH. Neumann DA. Kinesiology of the Musculoskeletal System. Magee DJ. Orthopedic Physical Assessment. Brukner & Khan Clinical Sports Medicine.

RECORD: Mulder's Click

ID: FOOT-TEST-02

Test: Mulder's click / squeeze

Purpose: Morton neuroma

Positive Finding: Painful click with dorsal-plantar compression + web squeeze

Sensitivity: Moderate

Specificity: Moderate — clinical diagnosis supported by US

Evidence: Common clinical test; confirm with imaging if invasive care planned

References: Standring S. Gray's Anatomy. Moore KL et al. Clinically Oriented Anatomy. Netter FH. Neumann DA. Kinesiology of the Musculoskeletal System. Magee DJ. Orthopedic Physical Assessment. Brukner & Khan Clinical Sports Medicine.

RECORD: Tinel at Tarsal Tunnel

ID: FOOT-TEST-03

Test: Tinel sign tarsal tunnel

Purpose: Tibial nerve entrapment

Positive Finding: Tingling into sole with percussion behind medial malleolus

Sensitivity: Limited alone

Specificity: Limited alone

Evidence: Part of cluster with neurodynamics and sensory testing

References: Standring S. Gray's Anatomy. Moore KL et al. Clinically Oriented Anatomy. Netter FH. Neumann DA. Kinesiology of the Musculoskeletal System. Magee DJ. Orthopedic Physical Assessment. Brukner & Khan Clinical Sports Medicine.

RECORD: Too-Many-Toes Sign

ID: FOOT-TEST-04

Test: Too-many-toes sign

Purpose: Hindfoot valgus / AAFD screening

Positive Finding: More than 1-2 toes visible lateral when viewed from behind

Sensitivity: Clinical screening

Specificity: N/A formal

Evidence: Classic PTTD/AAFD exam element with heel rise

References: Standring S. Gray's Anatomy. Moore KL et al. Clinically Oriented Anatomy. Netter FH. Neumann DA. Kinesiology of the Musculoskeletal System. Magee DJ. Orthopedic Physical Assessment. Brukner & Khan Clinical Sports Medicine.

RECORD: Single Heel Rise (Foot)

ID: FOOT-TEST-05

Test: Single heel rise for TP

Purpose: TP functional strength / staging PTTD

Positive Finding: Inability to invert/rise or early abort

Sensitivity: Clinically useful

Specificity: Pain may limit

Evidence: Core staging tool for PTTD

References: Standring S. Gray's Anatomy. Moore KL et al. Clinically Oriented Anatomy. Netter FH. Neumann DA. Kinesiology of the Musculoskeletal System. Magee DJ. Orthopedic Physical Assessment. Brukner & Khan Clinical Sports Medicine.

RECORD: Piano Key Test (Lisfranc)

ID: FOOT-TEST-06

Test: Piano key / midfoot stress

Purpose: Lisfranc instability suspicion

Positive Finding: Pain/mobility midfoot with dorsal-plantar MT translation

Sensitivity: Limited — high clinical suspicion needed

Specificity: Limited

Evidence: Any midfoot trauma with swelling/ecchymosis plantar → image WB

References: Standring S. Gray's Anatomy. Moore KL et al. Clinically Oriented Anatomy. Netter FH. Neumann DA. Kinesiology of the Musculoskeletal System. Magee DJ. Orthopedic Physical Assessment. Brukner & Khan Clinical Sports Medicine.

12. Pathologies

RECORD: Plantar Fasciopathy

ID: FOOT-PATH-01

Condition: Plantar fasciopathy

Epidemiology: Very common inferior heel pain; runners and middle-aged adults

Mechanism: Repetitive tensile overload at calcaneal enthesis continuum

Symptoms: First-step morning pain, medial calcaneal tubercle tenderness

Differential Diagnosis: Fat pad, Baxter nerve, calcaneal stress Fx, S1 radiculopathy

Clinical Tests: Windlass, palpation

Imaging: US thickening; MRI if atypical/stress Fx concern

Conservative Treatment: Load mod, stretching, calf capacity, orthoses, SWT selected; CSI judicious

Surgical Treatment: Rare partial release selected failures

Rehabilitation: 12+ weeks progressive often

Return to Sport: Pain-guided running build

Prognosis: Good majority with active care

References: Standring S. Gray's Anatomy. Moore KL et al. Clinically Oriented Anatomy. Netter FH. Neumann DA. Kinesiology of the Musculoskeletal System. Magee DJ. Orthopedic Physical Assessment. Brukner & Khan Clinical Sports Medicine.

RECORD: Morton's Neuroma

ID: FOOT-PATH-02

Condition: Morton's neuroma

Epidemiology: Women > men; 3-4 web most common

Mechanism: Perineural fibrosis interdigital nerve under transverse MT ligament

Symptoms: Forefoot burning, pebble-in-shoe, toe paresthesia

Differential Diagnosis: Metatarsalgia, plantar plate, stress Fx, bursitis

Clinical Tests: Mulder

Imaging: US/MRI

Conservative Treatment: Wide toe box, met pad, injection

Surgical Treatment: Neurectomy selected

Rehabilitation: Footwear + intrinsic dome

Return to Sport: When shoe/load tolerated

Prognosis: Variable; stump neuroma risk after surgery

References: Standring S. Gray's Anatomy. Moore KL et al. Clinically Oriented Anatomy. Netter FH. Neumann DA. Kinesiology of the Musculoskeletal System. Magee DJ. Orthopedic Physical Assessment. Brukner & Khan Clinical Sports Medicine.

RECORD: Hallux Valgus

ID: FOOT-PATH-03

Condition: Hallux valgus (bunion)

Epidemiology: Common; female predominance; footwear/genetic factors

Mechanism: Progressive 1st MTP valgus / MT varus with medial capsule failure

Symptoms: Medial eminence pain, shoe conflict, transfer metatarsalgia

Differential Diagnosis: Gout, hallux rigidus, sesamoid

Clinical Tests: ROM, alignment, callus map

Imaging: WB XR angles (HVA, IMA)

Conservative Treatment: Wide toe box, orthoses, pain mod — does not reverse deformity

Surgical Treatment: Osteotomy/soft tissue procedures individualized

Rehabilitation: Post-op protocols surgeon-specific; gait/1st ray function

Return to Sport: Months post-op

Prognosis: Surgery for pain/function not cosmetics alone ideally

References: Standring S. Gray's Anatomy. Moore KL et al. Clinically Oriented Anatomy. Netter FH. Neumann DA. Kinesiology of the Musculoskeletal System. Magee DJ. Orthopedic Physical Assessment. Brukner & Khan Clinical Sports Medicine.

RECORD: Hallux Rigidus

ID: FOOT-PATH-04

Condition: Hallux rigidus

Epidemiology: Common 1st MTP OA

Mechanism: Degenerative 1st MTP cartilage loss ± osteophytes

Symptoms: DF pain/stiffness, dorsal osteophyte irritation

Differential Diagnosis: Turf toe, gout, sesamoid

Clinical Tests: Grind, limited DF

Imaging: WB XR

Conservative Treatment: Rocker shoe, carbon insert, injection, activity mod

Surgical Treatment: Cheilectomy, fusion, implant selected

Rehabilitation: Protect DF extremes early post-op per procedure

Return to Sport: Fusion limits some sports — counsel

Prognosis: Fusion reliable pain relief

References: Standring S. Gray's Anatomy. Moore KL et al. Clinically Oriented Anatomy. Netter FH. Neumann DA. Kinesiology of the Musculoskeletal System. Magee DJ. Orthopedic Physical Assessment. Brukner & Khan Clinical Sports Medicine.

RECORD: Metatarsalgia

ID: FOOT-PATH-05

Condition: Metatarsalgia (mechanical)

Epidemiology: Common forefoot overload

Mechanism: Excess pressure MT heads — long 2nd MT, cavus, HV transfer, fat pad atrophy

Symptoms: Plantar MT head pain with WB

Differential Diagnosis: Neuroma, plantar plate, Freiberg, stress Fx

Clinical Tests: Palpation, drawer of toe

Imaging: XR WB; MRI if Fx/plate

Conservative Treatment: Met pads, footwear, intrinsic strength, activity mod

Surgical Treatment: Address deformity drivers selected

Rehabilitation: Load redistribution

Return to Sport: Graded

Prognosis: Good if drivers addressed

References: Standring S. Gray's Anatomy. Moore KL et al. Clinically Oriented Anatomy. Netter FH. Neumann DA. Kinesiology of the Musculoskeletal System. Magee DJ. Orthopedic Physical Assessment. Brukner & Khan Clinical Sports Medicine.

RECORD: Lisfranc Injury

ID: FOOT-PATH-06

Condition: Lisfranc injury

Epidemiology: Often missed in ED; sports and trauma

Mechanism: Axial PF load / twist

Symptoms: Midfoot pain/swelling, plantar ecchymosis classic, inability to push off

Differential Diagnosis: Ankle sprain mislabel, MT base fracture

Clinical Tests: Piano key; high suspicion

Imaging: WB XR bilateral; CT/MRI

Conservative Treatment: Stable nondisplaced — protected NWB often

Surgical Treatment: ORIF or primary arthrodesis if unstable

Rehabilitation: Prolonged; delayed run

Return to Sport: Often 4-6+ months

Prognosis: Missed → midfoot collapse/OA

References: Standring S. Gray's Anatomy. Moore KL et al. Clinically Oriented Anatomy. Netter FH. Neumann DA. Kinesiology of the Musculoskeletal System. Magee DJ. Orthopedic Physical Assessment. Brukner & Khan Clinical Sports Medicine.

RECORD: Turf Toe

ID: FOOT-PATH-07

Condition: Turf toe (1st MTP plantar plate/capsule sprain)

Epidemiology: Football/artificial surfaces classic

Mechanism: Hyperextension 1st MTP

Symptoms: 1st MTP pain, swelling, push-off weakness

Differential Diagnosis: Sesamoid fracture, HV

Clinical Tests: DF stress pain, weakened toe-off

Imaging: XR; MRI grade soft tissue

Conservative Treatment: Tapingspica, rigid insert, graded ROM grade I-II

Surgical Treatment: Grade III selected repair

Rehabilitation: Protect hyperextension; progressive push-off

Return to Sport: Grade-dependent weeks–months

Prognosis: Residual stiffness/pain if severe

References: Standring S. Gray's Anatomy. Moore KL et al. Clinically Oriented Anatomy. Netter FH. Neumann DA. Kinesiology of the Musculoskeletal System. Magee DJ. Orthopedic Physical Assessment. Brukner & Khan Clinical Sports Medicine.

RECORD: Metatarsal / Navicular Stress Fracture

ID: FOOT-PATH-08

Condition: Foot bone stress injuries

Epidemiology: Runners; navicular high-risk; MT2-3 common

Mechanism: Load > remodeling; energy availability

Symptoms: Focal bony pain, hop pain

Differential Diagnosis: Neuroma, synovitis

Clinical Tests: Point tenderness, hop

Imaging: MRI gold early

Conservative Treatment: Unload risk-stratified; RED-S screen

Surgical Treatment: Navicular/selected 5th MT Jones may need fixation

Rehabilitation: Graded return run

Return to Sport: Site-dependent

Prognosis: Guarded high-risk sites

References: Standring S. Gray's Anatomy. Moore KL et al. Clinically Oriented Anatomy. Netter FH. Neumann DA. Kinesiology of the Musculoskeletal System. Magee DJ. Orthopedic Physical Assessment. Brukner & Khan Clinical Sports Medicine.

RECORD: Pes Planus / Cavus

ID: FOOT-PATH-09

Condition: Pes planus and pes cavus foot types

Epidemiology: Planus common flexible childhood; cavus neurologic workup if progressive

Mechanism: Planus: ligamentous laxity/PTTD; Cavus: plantarflexed 1st ray / neurologic

Symptoms: Planus: medial strain; Cavus: lateral overload, ankle INV sprains, metatarsalgia

Differential Diagnosis: Coalition (rigid planus youth); Charcot-Marie-Tooth for cavus

Clinical Tests: Coleman block, heel rise, neurologic exam cavus

Imaging: WB XR; MRI/neuro studies as indicated

Conservative Treatment: Orthoses matched to type; strength; footwear

Surgical Treatment: Realignment selected symptomatic

Rehabilitation: Type-specific loading

Return to Sport: Individual

Prognosis: Depends on cause

References: Standring S. Gray's Anatomy. Moore KL et al. Clinically Oriented Anatomy. Netter FH. Neumann DA. Kinesiology of the Musculoskeletal System. Magee DJ. Orthopedic Physical Assessment. Brukner & Khan Clinical Sports Medicine.

RECORD: Sesamoiditis / Fat Pad Contusion

ID: FOOT-PATH-10

Condition: Sesamoiditis and heel fat pad contusion

Epidemiology: Runners, dancers; older adults fat pad

Mechanism: Repetitive plantar overload

Symptoms: Plantar 1st MTP or central heel pain — fat pad worse with hard heel strike

Differential Diagnosis: Fasciopathy (more medial calcaneal), Fx sesamoid

Clinical Tests: Palpation map; windlass less for fat pad

Imaging: XR/MRI sesamoid

Conservative Treatment: Offload pads, footwear, activity mod

Surgical Treatment: Rare sesamoidectomy risks

Rehabilitation: Graded load

Return to Sport: When offload strategies allow training

Prognosis: Good with load care

References: Standring S. Gray's Anatomy. Moore KL et al. Clinically Oriented Anatomy. Netter FH. Neumann DA. Kinesiology of the Musculoskeletal System. Magee DJ. Orthopedic Physical Assessment. Brukner & Khan Clinical Sports Medicine.

13. Rehabilitation

RECORD: Plantar Fasciopathy Rehab

ID: FOOT-REHAB-01

Condition: Plantar fasciopathy

Acute Phase: Reduce provocative impact; maintain fitness cross-train

Protection: Supportive shoes; avoid barefoot on hard floors initially if irritable

Mobility: Plantar fascia and calf stretches

Strength: Heavy slow calf + intrinsic foot

Balance: SLS arch control

Plyometrics: Late

Running Progression: Pain-guided walk-run

Cutting: After linear tolerance

Sport-Specific Drills: Graded

Return-to-Play Criteria: First-step pain minimal; training loads stable 2+ weeks

References: Standring S. Gray's Anatomy. Moore KL et al. Clinically Oriented Anatomy. Netter FH. Neumann DA. Kinesiology of the Musculoskeletal System. Magee DJ. Orthopedic Physical Assessment. Brukner & Khan Clinical Sports Medicine.

RECORD: Turf Toe Rehab

ID: FOOT-REHAB-02

Condition: Turf toe

Acute Phase: Protect hyperextension (boot/rigid insert)

Protection: Toe spica taping for return

Mobility: Graded MTP ROM after protection

Strength: FHL/FHB isometrics → isotonic

Balance: When WB comfortable

Plyometrics: Delayed push-off drills

Running Progression: After pain-free walk/push-off

Cutting: Late

Sport-Specific Drills: Position-specific with taping

Return-to-Play Criteria: Near-full DF without pain; push-off power; tape/insert plan

References: Standring S. Gray's Anatomy. Moore KL et al. Clinically Oriented Anatomy. Netter FH. Neumann DA. Kinesiology of the Musculoskeletal System. Magee DJ. Orthopedic Physical Assessment. Brukner & Khan Clinical Sports Medicine.

RECORD: Lisfranc Rehab Principles

ID: FOOT-REHAB-03

Condition: Lisfranc injury

Acute Phase: NWB often until stability confirmed

Protection: Boot/cast per ortho

Mobility: After WB allowed — midfoot careful

Strength: Proximal + progressive foot intrinsics

Balance: Delayed

Plyometrics: Late

Running Progression: Often 3-6+ months

Cutting: Last

Sport-Specific Drills: Graded midfoot load

Return-to-Play Criteria: Surgeon clearance, hop, pain-free push-off, imaging as indicated

References: Standring S. Gray's Anatomy. Moore KL et al. Clinically Oriented Anatomy. Netter FH. Neumann DA. Kinesiology of the Musculoskeletal System. Magee DJ. Orthopedic Physical Assessment. Brukner & Khan Clinical Sports Medicine.

RECORD: Metatarsal Stress Fracture Rehab

ID: FOOT-REHAB-04

Condition: Metatarsal stress fracture

Acute Phase: Unload impact

Protection: Boot/ crutches per site

Mobility: Maintain ankle/hip

Strength: Cross-train non-impact

Balance: When WB ok

Plyometrics: After bone healing window

Running Progression: Walk-run after clinical clearance

Cutting: After continuous run

Sport-Specific Drills: Graded

Return-to-Play Criteria: Pain-free hop and run volume build without delayed pain

References: Standring S. Gray's Anatomy. Moore KL et al. Clinically Oriented Anatomy. Netter FH. Neumann DA. Kinesiology of the Musculoskeletal System. Magee DJ. Orthopedic Physical Assessment. Brukner & Khan Clinical Sports Medicine.

RECORD: Morton Neuroma Rehab

ID: FOOT-REHAB-05

Condition: Morton's neuroma

Acute Phase: Wide toe box immediately

Protection: Metatarsal pad proximal to heads

Mobility: Calf/ankle if limited DF increases forefoot load

Strength: Intrinsic dome

Balance: SLS

Plyometrics: As tolerated

Running Progression: If footwear/pad controls symptoms

Cutting: As tolerated

Sport-Specific Drills: Footwear critical

Return-to-Play Criteria: Symptom-controlled sport volumes; injection/surgery pathway if failed

References: Standring S. Gray's Anatomy. Moore KL et al. Clinically Oriented Anatomy. Netter FH. Neumann DA. Kinesiology of the Musculoskeletal System. Magee DJ. Orthopedic Physical Assessment. Brukner & Khan Clinical Sports Medicine.

14. Evidence

RECORD: Evidence — Foot Module

ID: FOOT-EVID-01

Anatomy_Biomechanics: Standring S. Gray's Anatomy. Moore KL et al. Clinically Oriented Anatomy. Netter FH. Neumann DA. Kinesiology of the Musculoskeletal System. Magee DJ. Orthopedic Physical Assessment. Brukner & Khan Clinical Sports Medicine. Hicks windlass mechanism.

Plantar_Fasciopathy: Active loading and education first-line themes in clinical guidelines; CSI not first-line long-term.

Lisfranc: High index of suspicion; WB imaging; unstable injuries surgical.

Diabetic_Foot: Red-hot swollen foot — Charcot vs infection — urgent specialist care (educational red flag).

AI_Note: Angles, Sn/Sp, timelines approximate — verify individually.

References: Standring S. Gray's Anatomy. Moore KL et al. Clinically Oriented Anatomy. Netter FH. Neumann DA. Kinesiology of the Musculoskeletal System. Magee DJ. Orthopedic Physical Assessment. Brukner & Khan Clinical Sports Medicine.